

Forecasting Canyon Lake Storage: Why ARIMA Outperforms for Water Supply Planning

Amanda Marrero

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Contents

Introduction	2
Setup & Libraries	2
Data Collection & Preparation	3
Full Series Time Plot	4
Train / Test Split	5
Exploratory Analysis (Training Data Only)	6
ACF / PACF	6
STL Decomposition	6
Transformation & Stationarity Checks	7
Model Fitting	8
Model Summaries	8
Residual Diagnostics	9
Ljung-Box Test	12
Forecasting	12
Forecast Plot	13
Forecast Accuracy	13
Model Selection & Justification	14
References	14

Introduction

Canyon Lake, located in Comal County, Texas, serves as a primary surface water source for communities across the greater Central Texas region. The Guadalupe- Blanco River Authority (GBRA) administers the water for wholesale, and the reservoir's conservation levels directly govern water availability for municipal supply contracts throughout the Guadalupe River Basin. The city of Kyle currently receives 54% of its total permitted water supply from Canyon Reservoir, but contractually wholesale customers are subject to curtailments across six drought response stages, each triggered by declining reservoir levels. Although the reservoir has a capacity of 123 billion gallons, persistent drought conditions have significantly reduced current storage levels, increasing the risk of supply restrictions (City of Kyle, 2025). The city of Kyle has experienced significant growth over the last 25 years, increasing from 5,000 residents in 2000 to 50,000 in 2020, with projections reaching 150,000 by 2080. Kyle remained under Stage 3 drought restrictions through 2025, and as the city pursues continued economic expansion, accurate long-term water supply forecasting has become an operational necessity (KXAN Austin, 2025). Time series forecasting is a well-established quantitative tool for resource planning, with applications ranging from energy demand to reservoir storage management. Das et al. (2023) demonstrated that time series models applied to Texas reservoirs provide water planners improved ability to make informed decisions about water usage, minimizing impacts to local communities and businesses. This project applies four forecasting methodologies — Exponential Smoothing (ETS), ARIMA, Time Series Linear Regression (TSLM), and an Ensemble average — to monthly Canyon Lake storage data spanning January 2010 through December 2024, following the modeling framework described in Hyndman & Athanasopoulos (2021). The goal is to identify the model best suited for supporting water supply planning decisions in a rapidly urbanizing region under recurring drought pressure.

Setup & Libraries

```
library(tidyverse)
library(fpp3)
library(patchwork)
```

```
# Canyon Lake color palette - water/canyon theme
model_colors <- c(
  "ETS_Auto"   = "#2E86AB",    # steel blue
  "ARIMA_Auto" = "#A23B3B",    # clay red + selected model gets warmest color
  "TSLM"       = "#3D7A5C",    # canyon green
  "Ensemble"   = "#7B6FA0"     # muted purple
)

model_labels <- c(
  "ETS_Auto"   = "ETS (Automatic)",
  "ARIMA_Auto" = "ARIMA (Automatic)",
  "TSLM"       = "TSLM (Linear Regression)"
)

# Theme
theme_canyon <- function(accent_color = "#2E86AB") {
  theme_minimal() +
  theme(
    plot.title       = element_text(size = 11, face = "bold",
                                     color = "#1a3a2a"),
    plot.subtitle    = element_text(size = 9,   color = "#555555"),
    plot.caption     = element_text(size = 8,   color = "gray50", hjust = 0),
```

```

    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "#F7FAFA", color = NA),
    panel.grid.major = element_line(color = "#DDE8E8", linewidth = 0.4),
    panel.grid.minor = element_blank(),
    axis.title       = element_text(size = 9, color = "#333333"),
    axis.text        = element_text(size = 8, color = "#555555"),
    strip.text       = element_text(face = "bold", color = "white"),
    strip.background = element_rect(fill = accent_color),
    legend.position  = "bottom",
    legend.text      = element_text(size = 8),
    legend.title     = element_text(size = 8)
  )
}

theme_set(theme_canyon())

```

Data Collection & Preparation

```

# Read raw lines to identify header rows
raw_lines <- readLines("https://www.waterdatafortexas.org/reservoirs/individual/canyon.csv")

canyon_raw <- read.csv(
  "https://www.waterdatafortexas.org/reservoirs/individual/canyon.csv",
  skip      = 54,
  header    = TRUE,
  comment.char = ""
)

# Aggregate to monthly
canyon_monthly <- canyon_raw %>%
  mutate(date = as.Date(date),
         month = yearmonth(date)) %>%
  group_by(month) %>%
  summarise(
    percent_full      = mean(percent_full, na.rm = TRUE),
    conservation_storage = mean(conservation_storage, na.rm = TRUE)
  ) %>%
  as_tsibble(index = month)

# Filter to study window: Jan 2010 - Dec 2024
canyon_monthly <- canyon_monthly %>%
  filter(month >= yearmonth("2010 Jan") &
         month <= yearmonth("2024 Dec"))

# Validation checks
nrow(canyon_monthly) # should be 180

```

```
## [1] 180
```

```
range(canyon_monthly$month)
```

```
## <yearmonth[2]>  
## [1] "2010 Jan" "2024 Dec"
```

```
canyon_monthly %>% scan_gaps()
```

```
## # A tsibble: 0 x 1 [?]  
## # i 1 variable: month <mth>
```

Full Series Time Plot

```
canyon_monthly %>%  
  autoplot(percent_full, linewidth = 1.1) +  
  geom_vline(xintercept = yearmonth("2022 Jan"),  
            linetype = "dashed", color = "#7B6FA0", linewidth = 0.8) +  
  annotate("text", x = yearmonth("2018 Jun"), y = 85,  
          label = "Training", color = "#3D7A5C", size = 3) +  
  annotate("text", x = yearmonth("2023 Jun"), y = 85,  
          label = "Test", color = "#A23B3B", size = 3) +  
  labs(  
    title = "Canyon Lake: Percent Full (2010-2024)",  
    subtitle = "Dashed line indicates train/test split | Training: Jan 2010-Dec  
2021 | Test: Jan 2022-Dec 2024",  
    y = "Percent Full (%)",  
    x = NULL  
  )
```

Canyon Lake: Percent Full (2010–2024)

Dashed line indicates train/test split | Training: Jan 2010–Dec 2021 | Test: Jan 2022–Dec 2024



Train / Test Split

```
train_data <- canyon_monthly %>%  
  filter(month <= yearmonth("2021 Dec")) # 144 months (80%)  
  
test_data <- canyon_monthly %>%  
  filter(month > yearmonth("2021 Dec")) # 36 months (20%)  
  
cat("Training observations:", nrow(train_data), "\n")
```

```
## Training observations: 144
```

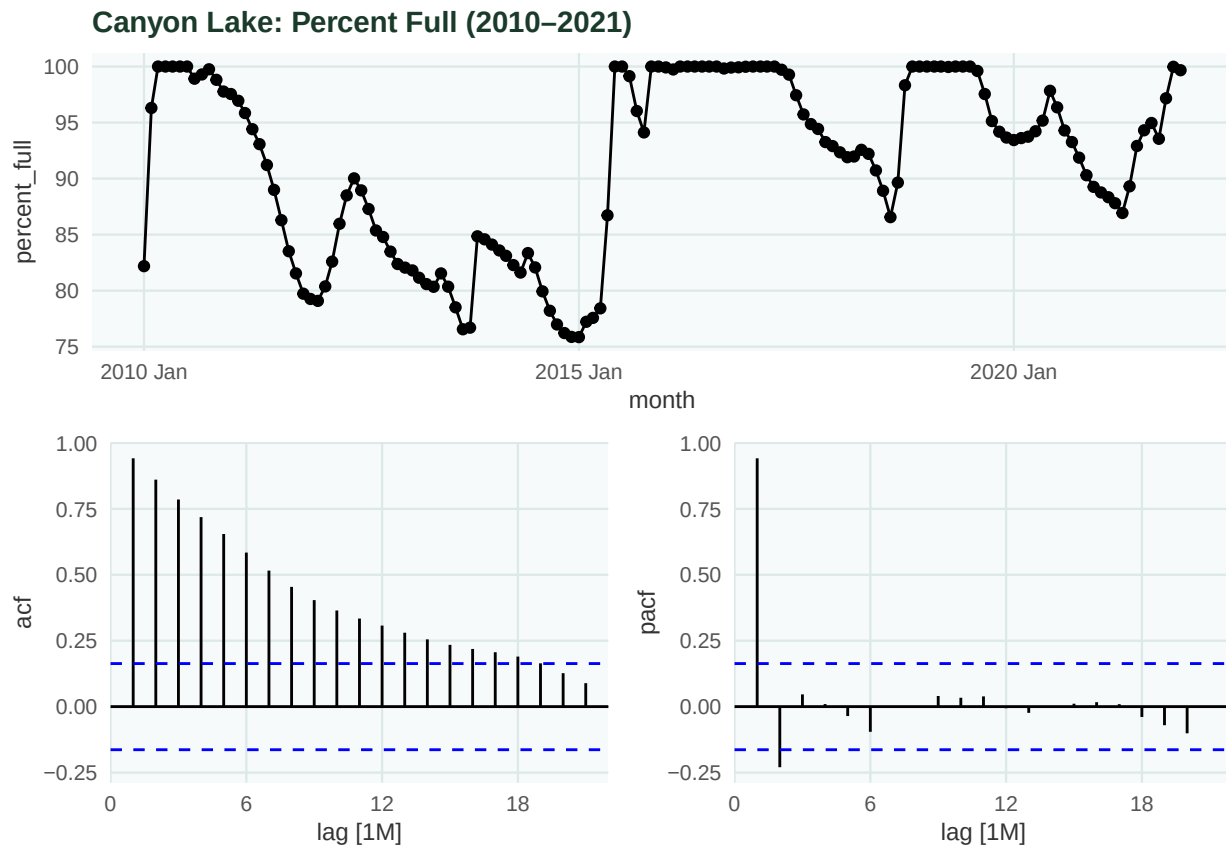
```
cat("Test observations: ", nrow(test_data), "\n")
```

```
## Test observations: 36
```

Exploratory Analysis (Training Data Only)

ACF / PACF

```
train_data %>%  
  gg_tsdisplay(percent_full, plot_type = "partial") +  
  labs(title = "Canyon Lake: Percent Full (2010-2021)"  
    )
```

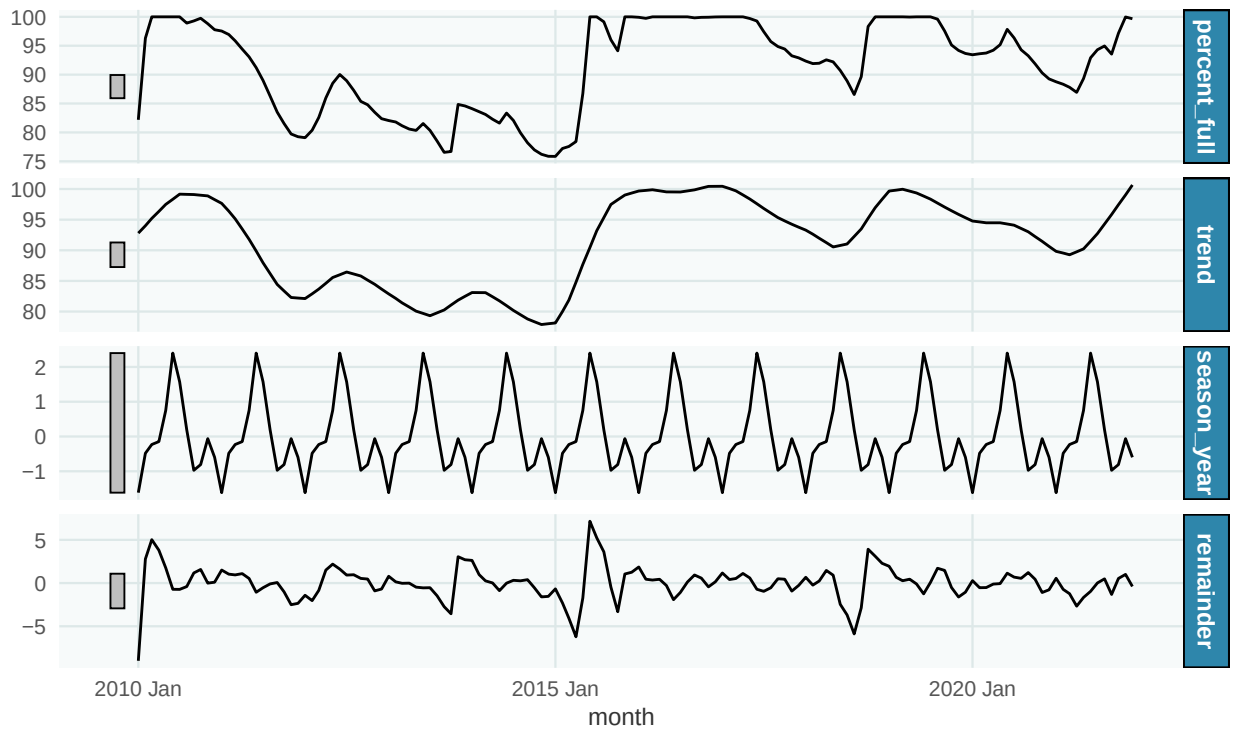


STL Decomposition

```
train_data %>%  
  model(STL(percent_full ~ trend(window = 13) +  
            season(window = "periodic"))) %>%  
  components() %>%  
  autoplot() +  
  labs(  
    title = "STL Decomposition - Canyon Lake Percent Full",  
    subtitle = "Training data: Jan 2010 - Dec 2021",  
    caption = "Source: Texas Water Data (waterdatafortexas.org)"  
  )
```

STL Decomposition — Canyon Lake Percent Full

Training data: Jan 2010 – Dec 2021



Source: Texas Water Data (waterdatafortexas.org)

Transformation & Stationarity Checks

```
# Box-Cox lambda (assessed but not applied due to bounded variable)
lambda <- train_data %>%
  features(percent_full, features = guerrero) %>%
  pull(lambda_guerrero)
cat("Guerrero lambda:", round(lambda, 3), "\n")
```

```
## Guerrero lambda: 0.516
```

```
# Seasonal differencing
train_data %>% features(percent_full, unitroot_nsdiffs)
```

```
## # A tibble: 1 x 1
##   nsdifs
##   <int>
## 1     0
```

```
# Non-seasonal differencing after seasonal diff
train_data %>% features(difference(percent_full, 12), unitroot_nsdiffs)
```

```
## # A tibble: 1 x 1
```

```
## ndiffs
## <int>
## 1 0
```

Model Fitting

```
models <- train_data %>%
  model(
    ETS_Auto = ETS(percent_full),
    ARIMA_Auto = ARIMA(percent_full),
    TSLM = TSLM(percent_full ~ trend() + season()),
    Ensemble = (ETS(percent_full) + ARIMA(percent_full) +
                TSLM(percent_full ~ trend() + season())) / 3
  )
```

Model Summaries

```
report(models$ETS_Auto[[1]])
```

```
## Series: percent_full
## Model: ETS(M,Ad,N)
## Smoothing parameters:
##   alpha = 0.9998999
##   beta  = 0.4551355
##   phi   = 0.8
##
## Initial states:
##   l[0]   b[0]
## 68.87447 13.65154
##
## sigma^2: 6e-04
##
##      AIC      AICc      BIC
## 961.4493 962.0624 979.2682
```

```
report(models$ARIMA_Auto[[1]])
```

```
## Series: percent_full
## Model: ARIMA(0,1,1)
##
## Coefficients:
##      ma1
##      0.5883
## s.e. 0.0824
##
## sigma^2 estimated as 4.751: log likelihood=-314.04
## AIC=632.09 AICc=632.17 BIC=638.01
```

```
report(models$TSLM[[1]])
```

```
## Series: percent_full
## Model: TSLM
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.4549  -5.9983  -0.2996   6.5164  12.7362
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   85.67585    2.44817   34.996 < 2e-16 ***
## trend()        0.05869    0.01546    3.797 0.000223 ***
## season()year2   1.15194    3.13717    0.367 0.714070
## season()year3   1.42267    3.13728    0.453 0.650960
## season()year4   1.49392    3.13747    0.476 0.634758
## season()year5   2.37826    3.13774    0.758 0.449841
## season()year6   4.00375    3.13808    1.276 0.204263
## season()year7   3.15510    3.13850    1.005 0.316612
## season()year8   1.74855    3.13900    0.557 0.578450
## season()year9   0.56015    3.13957    0.178 0.858671
## season()year10  0.75590    3.14022    0.241 0.810153
## season()year11  1.53278    3.14094    0.488 0.626364
## season()year12  1.02345    3.14174    0.326 0.745125
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.684 on 131 degrees of freedom
## Multiple R-squared:  0.1169, Adjusted R-squared:  0.03602
## F-statistic: 1.445 on 12 and 131 DF, p-value: 0.1535
```

Residual Diagnostics

```
# ARIMA residuals - styled (poster chart)
resid_plot <- models %>%
  select(ARIMA_Auto) %>%
  gg_tsresiduals(lag = 24) %>%
  invisible()
```

```

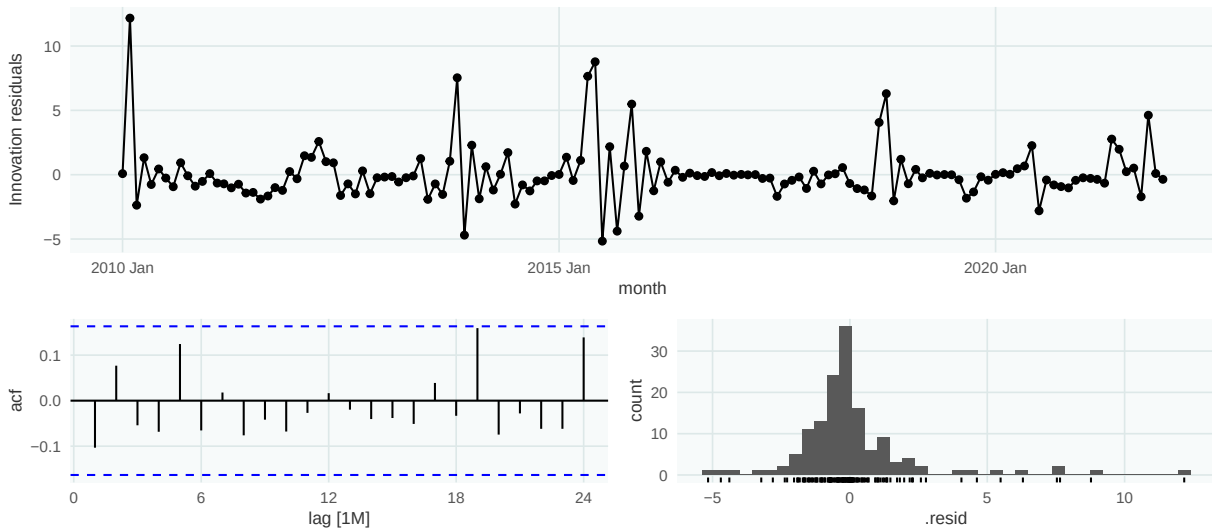
p1 <- resid_plot[[1]] + theme_canyon(accent_color = "#A23B3B") + labs(title = NULL)
p2 <- resid_plot[[2]] + theme_canyon(accent_color = "#A23B3B") + labs(title = NULL)
p3 <- resid_plot[[3]] + theme_canyon(accent_color = "#A23B3B") + labs(title = NULL)

(p1 / (p2 | p3)) +
  plot_layout(heights = c(1.5, 1)) +
  plot_annotation(
    title = "Residual Diagnostics \u2014 ARIMA (Automatic)",
    subtitle = "Innovations over time, ACF, and histogram",
    theme = theme(
      plot.title = element_text(size = 13, face = "bold", color = "#A23B3B"),
      plot.subtitle = element_text(size = 9, color = "#666666"),
      plot.background = element_rect(fill = "white", color = "#A23B3B", linewidth = 2)
    )
  )

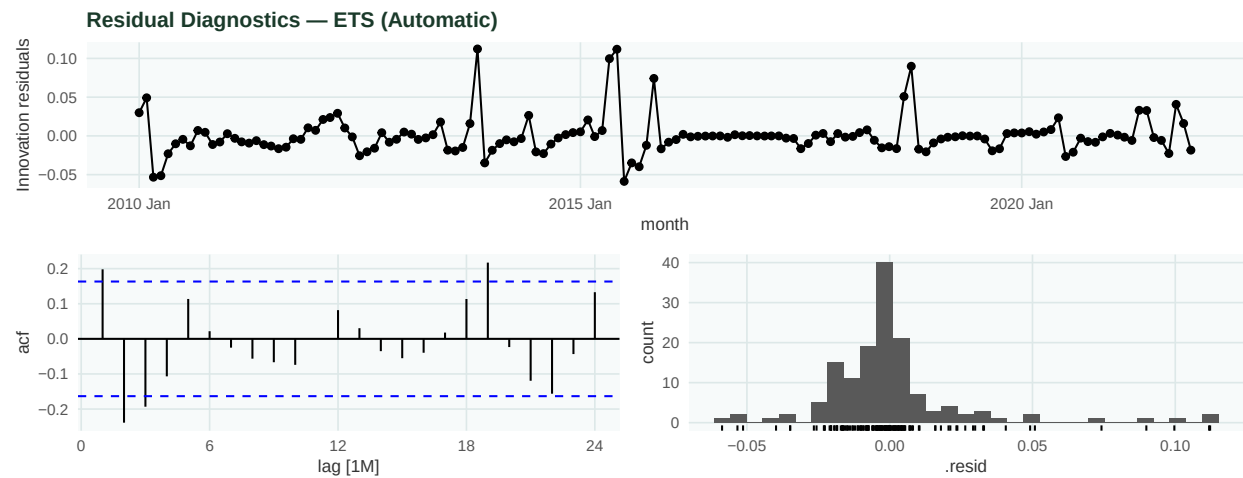
```

Residual Diagnostics — ARIMA (Automatic)

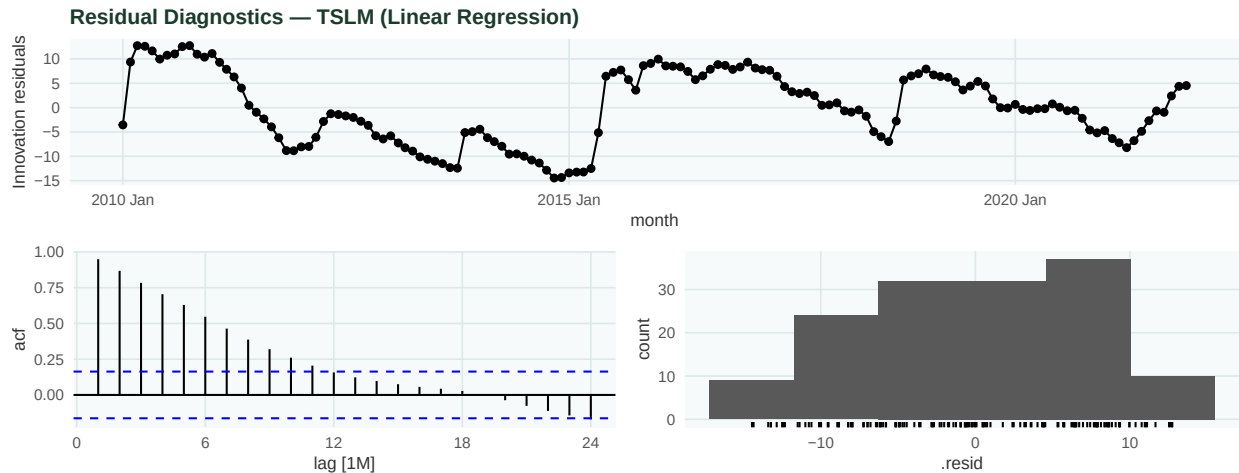
Innovations over time, ACF, and histogram



```
models %>%
  select(ETS_Auto) %>%
  gg_tsresiduals(lag = 24) +
  labs(title = "Residual Diagnostics - ETS (Automatic)")
```



```
models %>%
  select(TSLM) %>%
  gg_tsresiduals(lag = 24) +
  labs(title = "Residual Diagnostics - TSLM (Linear Regression)")
```



Ljung-Box Test

```
models_list <- c("ETS_Auto", "ARIMA_Auto", "TSLM")

ljung_results <- augment(models) %>%
  filter(.model %in% models_list) %>%
  group_by(.model) %>%
  features(.innov, ljung_box, lag = 24, dof = 0)

ljung_results %>%
  rename(Model = .model,
         `LB Statistic` = lb_stat,
         `p-value` = lb_pvalue) %>%
  mutate(
    `LB Statistic` = round(`LB Statistic`, digits = 2),
    `p-value` = round(`p-value`, digits = 3),
    `Residuals Clean?` = ifelse(`p-value` > 0.05, "Yes", "No"))
```

```
## # A tibble: 3 x 4
##   Model      'LB Statistic' 'p-value' 'Residuals Clean?'
##   <chr>          <dbl>      <dbl> <chr>
## 1 ARIMA_Auto      20.2      0.688 Yes
## 2 ETS_Auto       48.1      0.002 No
## 3 TSLM          627.        0     No
```

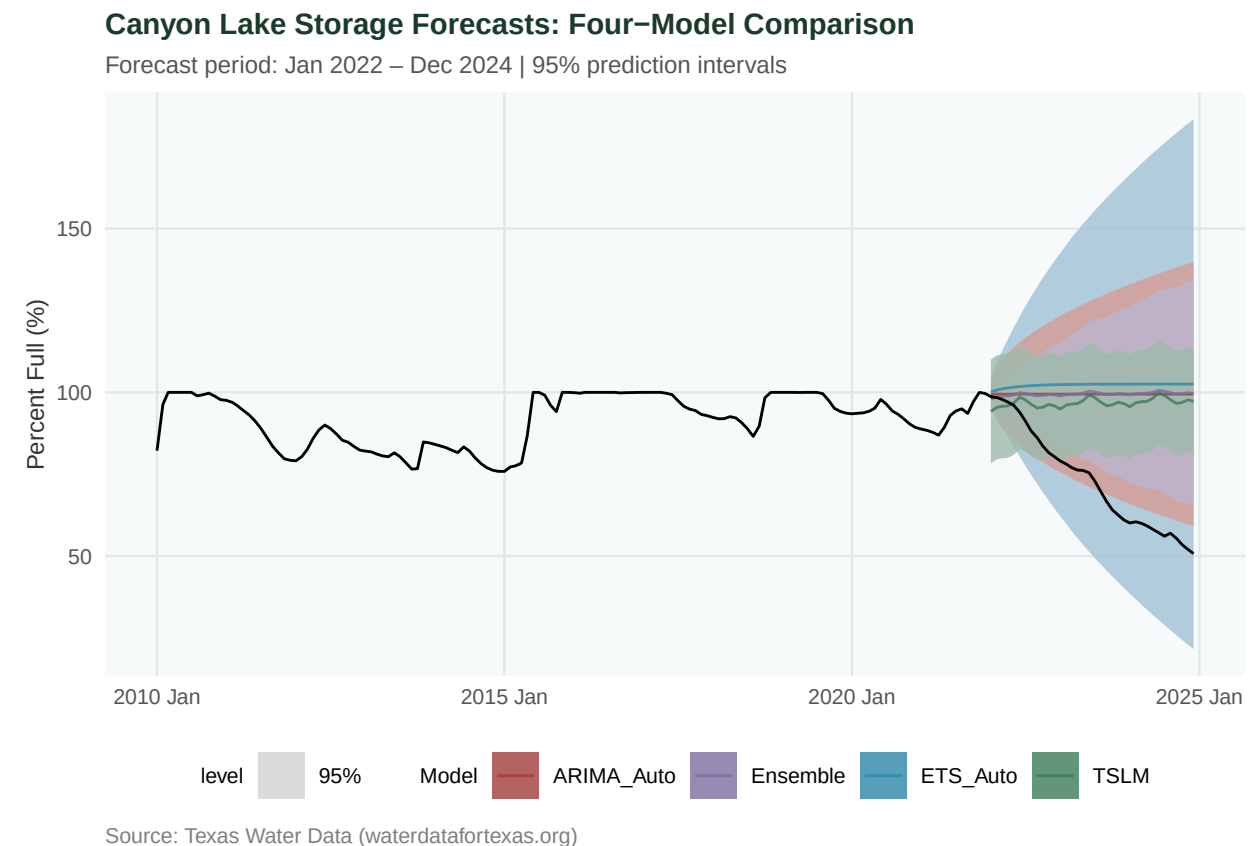
Interpretation: A p-value above 0.05 indicates residuals are consistent with white noise. ARIMA passes ($p = 0.688$); ETS and TSLM show significant residual autocorrelation.

Forecasting

```
fc <- forecast(models, h = 36)
```

Forecast Plot

```
fc %>%
  autoplot(canyon_monthly, level = 95, alpha = 0.8) +
  scale_color_manual(values = model_colors) +
  scale_fill_manual(values = model_colors) +
  labs(
    title = "Canyon Lake Storage Forecasts: Four-Model Comparison",
    subtitle = "Forecast period: Jan 2022 - Dec 2024 | 95% prediction intervals",
    caption = "Source: Texas Water Data (waterdatafortexas.org)",
    y = "Percent Full (%)", x = NULL, color = "Model", fill = "Model"
  )
```



Forecast Accuracy

```
accuracy_table <- fc %>%
  accuracy(test_data) %>%
  select(.model, RMSE, MAE, MAPE) %>%
  arrange(RMSE)

accuracy_table %>%
  rename(Model = .model) %>%
```

```
mutate(across(c(RMSE, MAE), ~ round(.x, 2)),
       MAPE = paste0(round(MAPE, 1), "%"))
```

```
## # A tibble: 4 x 4
##   Model      RMSE    MAE MAPE
##   <chr>      <dbl> <dbl> <chr>
## 1 TSLM       28.4   24.3 39%
## 2 ARIMA_Auto 30.4   26.4 42%
## 3 Ensemble   30.6   26.4 42.2%
## 4 ETS_Auto   33     29.1 46%
```

Model Selection & Justification

Although TSLM produced the lowest RMSE (28.5), its Ljung-Box p-value of approximately 0 indicates very autocorrelated residuals. This invalidates its prediction intervals, making it unreliable for contractual planning purposes where uncertainty bounds matter as much as point forecasts.

The ARIMA model is selected as the best model based on the combination of:

- Competitive forecast accuracy (RMSE = 30.4, MAE = 26.4, MAPE = 42.0%)
- Clean residuals confirmed by Ljung-Box test ($p = 0.688$)
- Statistically valid 95% prediction intervals suitable for drought-stage planning

For a water utility negotiating multi-year supply agreements under GBRA's tiered drought contingency framework, the lower bound of the forecast range is operationally critical. ARIMA is the only model that produces trustworthy uncertainty estimates.

References

- City of Kyle. (2022). *About Kyle water*. City of Kyle Water Utilities. <https://www.cityofkyle.gov/city-services/water-utilities/about-kyle-water/>
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- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
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- Water Data for Texas. (2024). *Canyon Lake reservoir storage data*. <https://www.waterdatafortexas.org/reservoirs/individual/canyon>